

		respectively. Thus, according to potential divider rule, $V_1 > V_2$
23	A	When the galvanometer has a current reading of zero, For minimum V_{output}, $R_{\text{output}} = 0 \Omega$, $V_{\text{output}} = 0 \text{ V}$ $V_x = V_{3.0 \Omega}$ $= \frac{R_{3.0}}{R_{\text{total}}} \times E = \frac{3.0}{3.0 + 9.0} \times 24$ $= 6.0 \text{ V}$
24	A	Current in the adjacent loops of the spring are parallel giving rise to attractive magnetic force between adjacent loops: spring is compressed